OilSim Report

OilSim: Oil Supply Chain Resilience Simulator

- **Course:** CS572 - Simulation and Modelling
- **Live Link:** http://34.121.56.100/
- **Repository:** github.com/AshwinG-23/OilSim
- **Supporting Documents:** README, Scenario and Feature Guide, Real-World Data Sources

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1. Project Description

What

OilSim is a Discrete-Event Simulation (DES) of India's crude oil import supply chain. It models how different oil-buying strategies perform when maritime routes are disrupted — by conflicts, sanctions, or fleet losses. Given a scenario (e.g., Strait of Hormuz blockade) and a strategy (e.g., always buy cheapest), the simulator runs the supply chain forward in time and reports how many hours the country ran short of oil, how much it spent per barrel, and how well demand was met.

Why

India imports roughly 85–90% of its crude oil, most of it passing through one of three maritime chokepoints: the Strait of Hormuz, the Red Sea/Suez Canal, or the Cape of Good Hope. A disruption at any of these points directly affects whether refineries keep running. The 2022 Russia–Ukraine war, the 2024 Houthi attacks on Red Sea shipping, and repeated Hormuz closure threats have all demonstrated that this is not a theoretical concern.

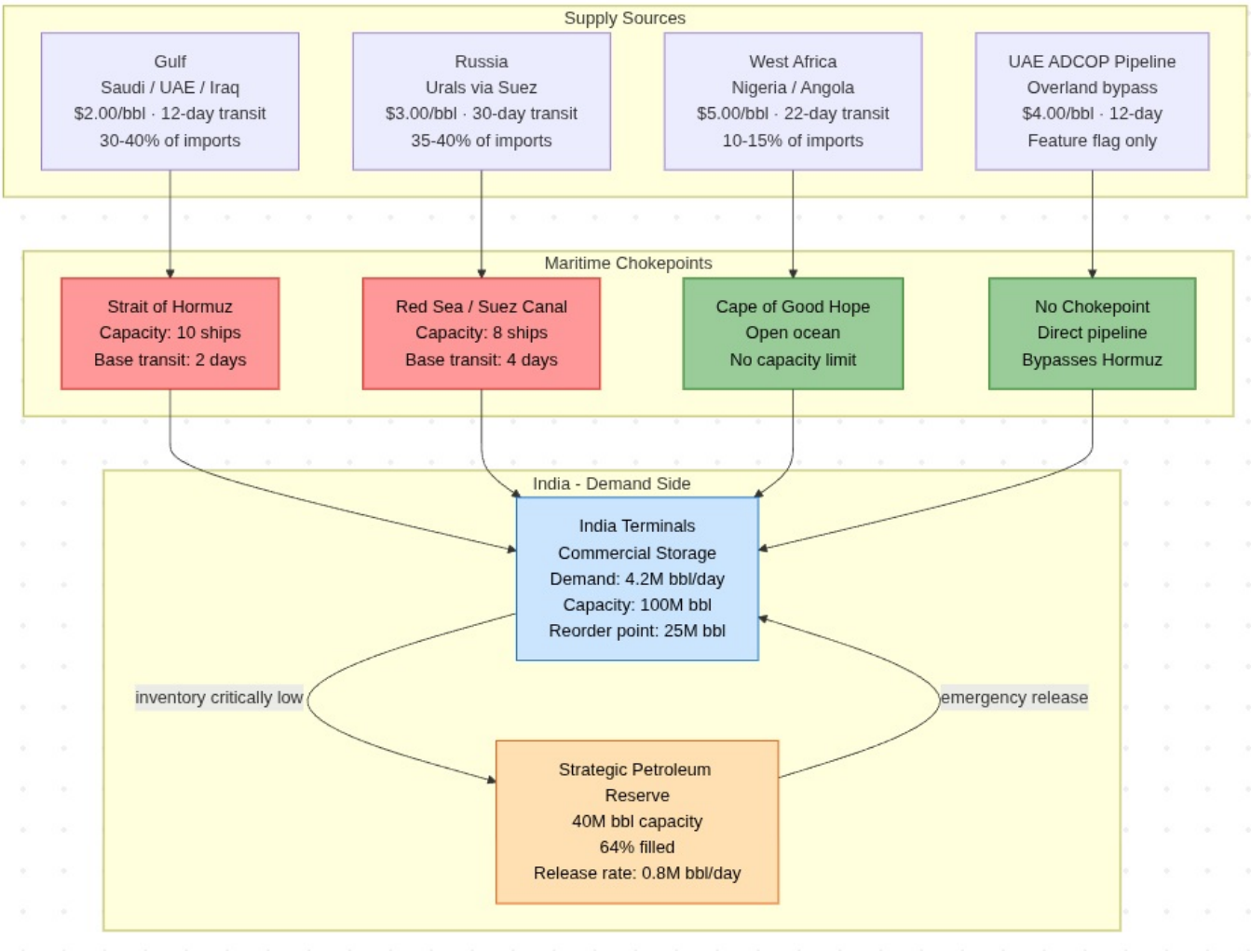
The core research question is: **Does the sourcing strategy India uses matter, and under which conditions does it matter most?** A cost-focused policy (always buy cheap Gulf oil) works perfectly in calm times but can cause severe shortages during a blockade. A resilience-focused policy (always route through Africa) avoids shortages but costs roughly 2.5× more per barrel even in peaceful periods. This simulation lets us quantify that trade-off.

How

The simulator is built using SimPy, a Python library for process-based discrete-event simulation. The entire physical system — tanker voyages, port inventory, chokepoint congestion, strategic reserves — is modelled as concurrent SimPy processes. Time advances in simulation hours; events such as a tanker completing a voyage or a chokepoint being disrupted are queued and processed in order.

The codebase is available on GitHub. A web-based dashboard (`web/app.py`) serves a live UI where scenarios can be configured and run interactively with Chart.js visualisations.

System Diagram



2. AI Tool Disclosure

Claude Code was used for a large share of the implementation work. I first created the project plan and design direction, then used Claude Code to help implement the simulator and the dashboard. Most of the dashboard was built with AI assistance and then refined through my feedback. The SimPy engine required several debugging passes, some of which I handled manually. A detailed interaction log is available at conversation_history.txt.

3. Methodology and Implementation

3.1 Simulation Architecture

The simulator has five physical components, each implemented as a SimPy process:

Component	What it models
Chokepoint	A maritime strait (Hormuz, Red Sea) with a capacity limit. Ships queue to transit. Disruptions lower throughput by reducing the `status` (0 = closed, 1 = normal).
Tanker / Fleet	120 tankers managed as a shared pool. Each tanker completes a full voyage cycle: load at source → transit chokepoints → unload at port → return.
Port	India's crude terminals. Demand (4.2M bbl/day) drains inventory continuously. When inventory falls below a reorder point, the ordering policy fires.
SPR	India's Strategic Petroleum Reserve (25.6M bbl at 64% fill). Releases automatically when commercial inventory drops to a critical threshold.
Disruption	Applies scenario events — lowering chokepoint status at a given start day for a given duration, cutting

Engine	the fleet, or shutting down a source.

The ordering loop runs every 24 simulation hours. If inventory is below the reorder point (25M bbl) and tankers are free, the policy places an order. This results in a tanker departing, spending time at sea, and delivering a cargo (1.5M bbl) back to the port inventory.

3.2 Simulation Parameters

All parameters are calibrated to real-world data (see Section 6 - References for sources).

Parameter	Value	Basis
Fleet size	120 tankers	India charters ~100–150 crude tankers at any time [1]
Cargo per tanker	1.5M bbl	Weighted average: VLCC (55%) + Suezmax (30%) + Aframax (15%) [2]
Daily demand	4.2M bbl/day	India's crude imports ~4.0–4.5M bbl/day [3]
Initial inventory	60M bbl	~14 days forward cover at commercial terminals [4]
Max storage	100M bbl	Combined commercial crude-tank capacity [4]
Reorder point	25M bbl	~6 days — triggers replenishment [4]
Gulf one-way transit	12 days	Gulf–India sea leg, 10–14 days typical [5]
Russia one-way transit	30 days	Baltic/Black Sea → Suez → India, 25–35 days [6]
Africa one-way transit	22 days	West Africa → Cape of Good Hope → India [7]
Gulf freight cost	$2.00/bbl	PPAC price build-up sheets [8]
Russia freight cost	$3.00/bbl	Reflects post-2022 freight + insurance [9]
Africa freight cost	$5.00/bbl	West Africa–Asia freight under normal conditions [10]
SPR capacity	40M bbl	Three caverns: Visakhapatnam, Padur, Mangaluru [11]
SPR fill level	64% (25.6M bbl)	Government disclosure, ~3.37M tonnes filled [11]
SPR release rate	0.8M bbl/day	Estimated based on pipeline throughput constraints [12]

Transit under disruption. When a chokepoint's status falls below 1.0, transit time increases inversely:

```
transit_time = base_transit / chokepoint_status + queue_delay
```

A status of 0.5 doubles transit time; status of 0.1 makes it 10× longer. Ships queue at the chokepoint resource in SimPy, so congestion compounds the delay as more ships pile up.

Disruption cost premium. When a tanker travels through a disrupted chokepoint, a war-risk multiplier is applied to the freight cost. At status 0.15, the multiplier reaches ×2.275 — consistent with insurance reports showing war-risk surcharges doubling or tripling base freight [13].

3.3 Supply Sources

Four sources are modelled. The UAE ADCOP pipeline starts disabled and only activates via a feature flag.

Source	Route	Chokepoints	Base Cost
Gulf (Saudi/UAE/Iraq)	Strait of Hormuz	Hormuz	$2.00/bbl
Russia (Urals via Suez)	Red Sea / Suez Canal	Red Sea	$3.00/bbl
West Africa (Nigeria/Angola)	Cape of Good Hope	None	$5.00/bbl

UAE ADCOP Pipeline	Overland bypass	None	$4.00/bbl

Russia's route in the model goes via the Red Sea, not Hormuz. This means Houthi attacks on the Red Sea affect Russia shipments but not Gulf shipments — which matches the real-world 2024 routing pattern.

3.4 Sourcing Strategies

Four strategies are implemented in `strategies/policy.py`. All share the same reorder trigger (inventory < 25M bbl). They differ in which source they pick.

Cost-Focused — buy from the cheapest route that is not severely disrupted. A chokepoint must fall below 30% capacity before this strategy considers it "disrupted." It defaults to Gulf in almost every situation.

Diversified — cycle through all non-disrupted sources in rotation (round-robin). It treats a chokepoint as disrupted at 70% capacity, so it spreads orders earlier. No single source dominates.

Resilient — always prefer routes with zero chokepoints, sorted by chokepoint count then cost. Africa is the default choice in every scenario. A chokepoint at 90% capacity is already treated as "disrupted." This strategy never routes through a strait unless every chokepoint-free source is unavailable.

Adaptive — observes the average health of all chokepoints every 24 hours and switches its disruption threshold dynamically:

- Average status ≥ 0.8 (calm): behaves like Cost-Focused
- Average status 0.5–0.8 (moderate stress): behaves like Diversified
- Average status < 0.5 (crisis): behaves like Resilient

This means Adaptive pays cheap Gulf prices during peace and automatically shifts to safe routes as conditions deteriorate, without any manual tuning.

3.5 Scenarios

Ten scenarios are defined in `config/params.py`:

Scenario	What happens
`no_disruption`	Normal operations — baseline
`short_conflict`	Hormuz at 50% for 30 days
`medium_blockade`	Hormuz at 20% for 90 days
`long_war`	Hormuz at 5% + Red Sea at 10% + 15 tankers lost
`all_out_war`	Hormuz 2% + Red Sea 2% + Russia sanctioned + 65 tankers lost
`cape_disruption`	Africa route closed + Hormuz 15% + Red Sea 20%
`houthi_red_sea`	Red Sea at 5% from day 0 for the full simulation
`russia_sanctions`	Russia supply shuts down from day 60 onward
`tanker_strike`	Hormuz at 30% for 60 days + 25 tankers lost on day 30
`stochastic_conflict`	Random disruptions drawn from a Poisson process

3.6 Feature Flags

Eight optional extensions can be toggled at runtime: tanker breakdowns, demand elasticity, seasonal weather, panic ordering (bullwhip effect), information delay, SPR auto-refill, freight rate dynamics, and UAE pipeline bypass. All are off by default; the results in this report use the default (no flags active).

3.7 Output Metrics

For each strategy × scenario combination, the simulator reports:

- **Shortage hours** — total hours where inventory hit zero
- **Demand Fulfillment Rate (DFR)** — fraction of demand that was actually met (1.0 = no shortage)
- **Average cost per barrel** — total transport spend ÷ total barrels delivered
- **Total spend** — cumulative freight cost over the simulation period
- **Fleet utilisation** — fraction of tankers in active service at any given time

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4. Results and Evaluation

All results below are from runs of 100 simulated days with 40 independent replications per cell. Full results are in `results.md`.

4.1 Results Table

Scenario	Strategy	Shortage (h)	DFR	$/bbl	Cost ($B)
no_disruption	cost_focused	0.0	1.000	2.00	0.76
no_disruption	diversified	0.0	1.000	3.32	1.26
no_disruption	resilient	0.0	1.000	5.00	1.87
no_disruption	adaptive	0.0	1.000	2.00	0.76
short_conflict	cost_focused	0.0	1.000	2.57	0.91
short_conflict	diversified	0.0	1.000	3.58	1.34
short_conflict	resilient	0.0	1.000	5.00	1.87
short_conflict	adaptive	0.0	1.000	2.71	1.03
medium_blockade	**cost_focused**	**402.8**	**0.832**	2.63	0.67
medium_blockade	diversified	0.0	1.000	3.86	1.44
medium_blockade	resilient	0.0	1.000	5.00	1.87
medium_blockade	adaptive	0.0	1.000	3.54	1.34
long_war	cost_focused	0.0	1.000	4.58	1.66
long_war	diversified	0.0	1.000	4.76	1.78
long_war	resilient	0.0	1.000	5.00	1.87
long_war	adaptive	0.0	1.000	4.66	1.72
all_out_war	cost_focused	136.0	0.943	5.00	1.42
all_out_war	diversified	136.0	0.943	5.00	1.42
all_out_war	resilient	136.0	0.943	5.00	1.42
all_out_war	adaptive	136.0	0.943	5.00	1.42
cape_disruption	cost_focused	1093.7	0.544	4.55	0.34
cape_disruption	**diversified**	**869.6**	**0.638**	5.24	0.68
cape_disruption	resilient	1094.2	0.544	4.55	0.34

cape_disruption	adaptive	1095.6	0.543	4.55	0.34
houthi_red_sea	cost_focused	0.0	1.000	2.00	0.76
houthi_red_sea	diversified	0.0	1.000	3.50	1.32
houthi_red_sea	resilient	0.0	1.000	5.00	1.87
houthi_red_sea	adaptive	0.0	1.000	3.50	1.32
russia_sanctions	cost_focused	0.0	1.000	2.00	0.76
russia_sanctions	diversified	0.0	1.000	3.38	1.28
russia_sanctions	resilient	0.0	1.000	5.00	1.87
russia_sanctions	adaptive	0.0	1.000	2.00	0.76
tanker_strike	**cost_focused**	**428.7**	**0.821**	3.24	0.75
tanker_strike	diversified	0.0	1.000	3.79	1.44
tanker_strike	resilient	0.0	1.000	5.00	1.87
tanker_strike	adaptive	0.0	1.000	3.44	1.27
stochastic_conflict	cost_focused	77.4	0.968	2.51	0.82
stochastic_conflict	diversified	0.0	1.000	3.70	1.39
stochastic_conflict	resilient	0.0	1.000	5.00	1.87
stochastic_conflict	**adaptive**	**0.0**	**1.000**	**3.07**	**1.15**

Bold rows highlight interesting results discussed below.

4.2 Key Findings

Finding 1: Cost-focused fails at sustained moderate disruption.

In `medium_blockade`, Hormuz drops to 20% capacity for 90 days. Cost-focused's disruption threshold is 0.3 — meaning it does not treat a 20% chokepoint as "disrupted." It keeps routing tankers through a congested strait, those tankers get stuck, the fleet backs up, and inventory runs out for 402 hours. Every other strategy avoids this by rerouting earlier.

The same pattern repeats in `tanker_strike`: a sudden loss of 25 tankers creates a throughput gap. Cost-focused has no slack in its routing, so it cannot absorb the shock. The other strategies, which maintain more spread across routes, keep enough tankers active to cover demand.

Finding 2: All strategies tie in all-out war — physical capacity sets a floor.

`all_out_war` closes every major chokepoint and destroys 65 tankers. The only usable route is Africa, but only 55 tankers remain. Africa's round trip takes ~24.5 days; at 1.5M bbl per tanker: throughput ≈ 55 × 1.5M / 24.5 = 3.37M bbl/day, which is below the 4.2M bbl/day demand. The 60M bbl reserve covers the gap for about 72 days, then fails for ~136 hours. No ordering policy can change this — the result is identical for all four strategies.

This is an important finding: **there is a physical lower bound on resilience**. Strategy only matters above that bound.

Finding 3: Cape disruption is the hardest scenario, and diversified wins by a margin.

`cape_disruption` closes Africa, disrupts Hormuz (15%), and disrupts Red Sea (20%). With Africa gone and no safe routes left, all strategies fall back to whatever is available. Cost-focused, resilient, and adaptive each pick the single cheapest/safest remaining source — Gulf — and use it exclusively. That gives them one route's worth of throughput through a slow, congested Hormuz.

Diversified does something different: it round-robins between Gulf and Russia. Using two degraded routes rather than one delivers about 70% more barrels (128M bbl total vs 75M bbl), reducing shortage hours by ~20%. The trade-off is higher cost ($5.24 vs $4.55/bbl) because Russia's route is also slow and expensive under disruption.

Finding 4: Adaptive is the best all-rounder.

Adaptive achieves DFR = 1.000 in 8 of 10 scenarios (same as resilient), but at much lower cost. In stochastic conflict — the scenario closest to real-world conditions where disruptions arrive unpredictably — adaptive pays $3.07/bbl with zero shortage, compared to resilient's $5.00/bbl (always pays Africa premium regardless of conditions) and cost-focused's 77 hours of shortage.

The cost advantage is clear in calm markets: adaptive charges $2.00/bbl in `no_disruption` and `russia_sanctions` — identical to cost-focused — because its threshold switches back to 0.3 when chokepoints recover.

Finding 5: Houthi and Russia sanctions don't stress the system under current fleet size.

Both `houthi_red_sea` and `russia_sanctions` result in DFR = 1.000 for all strategies. The 120-tanker fleet has enough capacity that even removing Russia entirely (35–40% of India's real-world import volume) can be compensated by Gulf + Africa shipments. This is consistent with observed market outcomes in 2022–24, where supply disruptions were absorbed through rerouting rather than shortages.

4.3 Cost vs Resilience Trade-off

The table below summarises the trade-off for a single scenario (`stochastic_conflict`) where conditions vary unpredictably — the closest to real-world operating conditions.

Strategy	Shortage (h)	$/bbl	Total Cost ($B)
cost_focused	77.4	2.51	0.82
diversified	0.0	3.70	1.39
resilient	0.0	5.00	1.87
adaptive	**0.0**	**3.07**	**1.15**

Adaptive eliminates shortage while costing $0.72B less per 100-day period than resilient. Extrapolated annually, that difference is roughly $2.6B — consistent with the real-world figure that Indian PSUs spend ~$8 billion per year on tanker charters [1], where optimised routing could deliver meaningful savings.

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5. Individual Contributions

Team Member	Contribution
Ashwin Gaikwad	Project design, simulation logic, debugging, evaluation, report writing, and deployment

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6. References

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